One RMU is equal to _____.

One RMU is equal to ≈45 mm (1.75 in). (ITSIMM, page 4-16)

What is the recommended minimum diameter of the toggle bolts used to mount plywood?

These toggle bolts should be a minimum of ≈6.3 mm (0.25) in diameter. (ITSIMM, page 4-12)

You are installing a plywood backboard in a telecommunications room (TR), and the plywood edges do not align with the studs. Which of the following should you use to attach the plywood to the framed wall?

When the plywood edges do not align with the studs, toggle bolts (butterfly bolts) may be required to attach the plywood to the framed wall. (ITSIMM, page 4-12)

Open-top cable supports should be installed to provide a minimum of _____ clearance above ceilings.

Where not specified by the AHJ, open-top cable supports should be installed to maintain a minimum of ≈75 mm (3 in) vertical clearance between the cabling and cable supports and elements such as ceiling tiles, electrical conduit, luminaires, and piping and ducting. (ITSIMM, page 4-63)

A third bend may be acceptable in a conduit segment without derating the conduit's capacity in certain circumstances. What are those circumstances? **Select all that apply**.

A third bend may be acceptable in a conduit segment without derating the conduit's capacity if any of the following exists: the run does not exceed ≈10 m (33 ft); the conduit size is increased to the next trade size; or one of the bends is located within ≈300 mm (12 in) of the cable feed end. The last exception only applies to placing operations where cable is pushed around the first bend. (ITSIMM, page 4-68)

- [x] The run does not exceed ≈10 m (33 ft).
- [x] The conduit size is increased to the next trade size.
- [x] One of the bends is located within ≈300 mm (12 in) of the cable feed end.

The term _____ is used to describe a connection that is intended to safely and effectively equalize the potential differences between two metallic items.

Bonding is the connection intended to safely and effectively equalize the potential differences between two metallic items (e.g., equipment rack and cable tray). (ITSIMM, page 8-1)

Which grounding system establishes a 0 volt (V) reference for ac electrical power systems?

The purpose of the ac electrical power grounding (earthing) electrode system is to establish a 0 V reference for ac electrical power systems, whether utility-provided or customer-derived. (ITSIMM, page 8-7)

What is the recommended maximum value for the bonding resistance between the two points on bonded surfaces?

The recommended maximum value for the bonding resistance between these two points is 0.1 ohms (100 milliohms). (ITSIMM, page 8-26)

When installing protection terminals, the NEC code allows the conductor to be a small as _____.

Some codes allow the conductor to be as small as 14 AWG [1.63 mm (0.064 in)]. (ITSIMM, page 8-34)

Bends of bonding conductors terminating at the primary bonding busbar (PBB) or secondary bonding busbar (SBB) shall have a minimum inside bend radius of _____.

Bends of bonding conductors terminating at the PBB or SBB shall have a minimum inside bend radius of ≈200 mm (8 in). (ITSIMM, page 8-16)

A BBC is an insulated, stranded copper conductor, with a minimum diameter size of _____.

A BBC is an insulated, stranded copper conductor, with a minimum diameter size of 6 AWG (4.1 mm [0.16 in]). (ITSIMM, page 8-17)

Which meeting should be held after the initial meeting and should include the cabling installation team and appropriate contractors?

- ✗ Initial Meeting
- ✓ Pre-Installation Meeting

The pre-installation meeting should include the cabling installation team and appropriate contractors. (ITSIMM, page 10-33)

What manual should contain the title of the project and the names and contact information for the architect or engineer, ICT systems designer, and ICT systems contractor?

O&M

An O&M manual should contain the title of the project and the names and contact information for the architect or engineer, ICT systems designer, and ICT systems contractor. Information for each product or system should be included identifying the specific products and components installed for the project. (ITSIMM, page 10-30)

When a pulse of light is sent down an optical fiber, part of the pulse is blocked by microscopic particles (called dopants) in the glass and scatters in all directions. What is this called?

- [x] Rayleigh scattering

An OTDR can be used to isolate problems in optical fiber cabling. (ITSIMM, page 6-80)

What does the troubleshooting process involve? Choose all that apply.

- [x] Verifying the problem
- [x] Repairing the problem
- [x] Isolating the source of the problem
- [x] Test the repaired system

The installer must identify, locate, and repair the problem and then test the repaired cabling system to ensure that it works correctly. (ITSIMM, page 6-26)

An inspection reveals that the splice is constricted. Which of the following are the most likely causes of this error? *Select all that apply*.

Constriction can result from many causes, including the feed being too small or the separation being too great. (ITSIMM, page 5-159)

Separation too great

Feed too small

Typically, fusion splice loss will be ____ dB or less.

Typically, fusion splice loss will be 0.1 dB or less. (ITSIMM, page 5-163)

Which of the following is the most common splice arrangement for optical fiber splices?

A butt arrangement is the most common for optical fiber. This arrangement has all the cables entering and exiting the enclosure from the same end. (ITSIMM, page 5-151)

When splicing fiber optic cabling, the maximum splice loss shall NOT exceed _____ dB regardless of the methodology used.

The maximum splice loss shall not exceed 0.3 dB, regardless of the methodology used. (ITSIMM, page 5-163)

Which of the following is the most common module used in permanent splices of copper cabling?

The straight module is the most common module used in permanent splices. (ITSIMM, page 5-139)

Which of the following termination processes requires no curing process?

Crimp termination requires no curing process. The fiber is secured within the connector by mechanical pressure within the connector body. The fiber within the ferrule is free floating. (ITSIMM, page 5-103)

Which of the following types of cable may require the installer to terminate a drain wire?

Screened/shielded cables typically include a drain wire and foil screen; both shall be terminated to the connector's screen/shield when present. (ITSIMM, page 5-78)

What color adapter should be used with 50/125 µm multimode laser enhanced optical fiber?

An aqua adapter should be with 50/125 µm multimode laser enhanced optical fiber. (ITSIMM, page 5-102, Table 5.5)

When terminating category 5e and higher cables, untwisting shall not exceed _____ from the termination point.

When terminating category 5e and higher performance cables, the cable pair twists shall be maintained as close as possible to the point of termination, but untwisting shall not exceed ≈12.7 mm (0.5 in) from the termination point. (ITSIMM, page 5-85)

When terminating category 3 cables, the cable pair twists must be maintained to within _____ from the termination point.

When terminating category 3 cables, the cable pair twists must be maintained to within ≈75 mm (3 in) from the termination point. (ITSIMM, page 5-85)

Which of the following connection processes uses a connector that contains a pre-polished fiber stub with index-matching gel?

Cleave and secure connectors contain a pre-polished fiber stub with index-matching gel. (ITSIMM, page 5-100)

What color adapter should be used with 62.5/125 µm multimode optical fiber?

A beige adapter should be used with 62.5/125 µm multimode optical fiber. (ITSIMM, page 5-102, Table 5.5)

Twisted-pair cable requires an _____ connection.

Twisted-pair cable requires an IDC connection. (ITSIMM, page 5-69)

Which of the following tools should be used to pull cable of significant weight from the bottom up?

A tugger is required to pull cable of significant weight from the bottom up. (ITSIMM, page 5-37)

The minimum inside bend radius for a four-strand optical fiber cable at rest is _____.

The minimum inside bend radius for a four-strand optical fiber cable at rest is ≈25 mm (1 in). (ITSIMM, page 5-7, Table 5.1)

Copper backbone cables shall have enough slack to

_____

Copper backbone cables shall have enough slack to reach the farthest corner of the backboard via the pathways. (ITSIMM, page 5-22)

When pulling backbone cable from the top down, the installation team should secure the cable _____.

When pulling backbone cable from the top down, the installation team should secure the cable at each floor. (ITSIMM, page 5-43)

The maximum pull force for 4-pair, 24 AWG [0.51 mm (0.020 in)] cable is _____.

The maximum pull force for 4-pair, 24 AWG [0.51 mm (0.020 in)] cable is 110 N (25 lbf). (ITSIMM, page 5-23)

When identifying the pull points for a horizontal cable run, it is important to ensure that the complete run does not exceed

_____.

When identifying the pull points for a horizontal cable run, it is important to ensure that the complete run does not exceed ≈90 m (295 ft). (ITSIMM, page 5-15)

Optical fiber cables shall have _____ of slack left in the work area to allow for termination.

Optical fiber cables shall have ≈1 m (3.3 ft) of slack left in the work area to allow for termination. (ITSIMM, page 5-22)

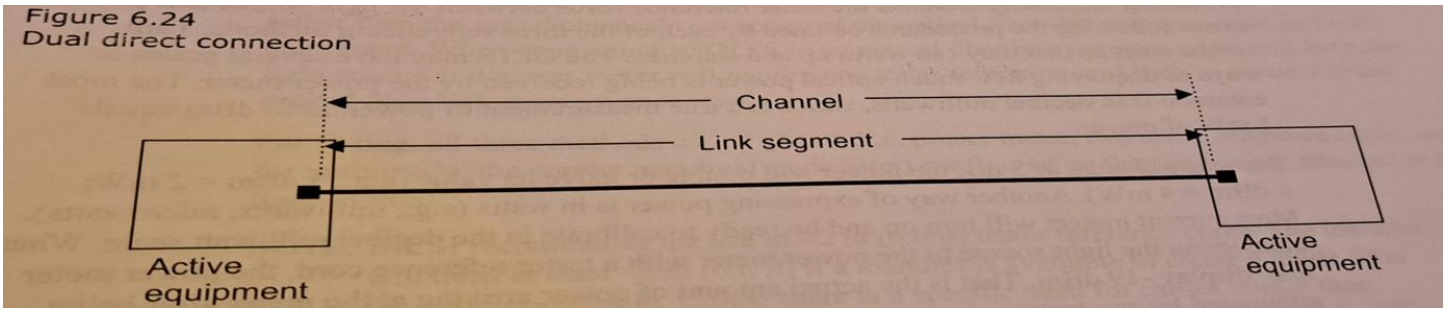


Figure 6.23
Single interconnection

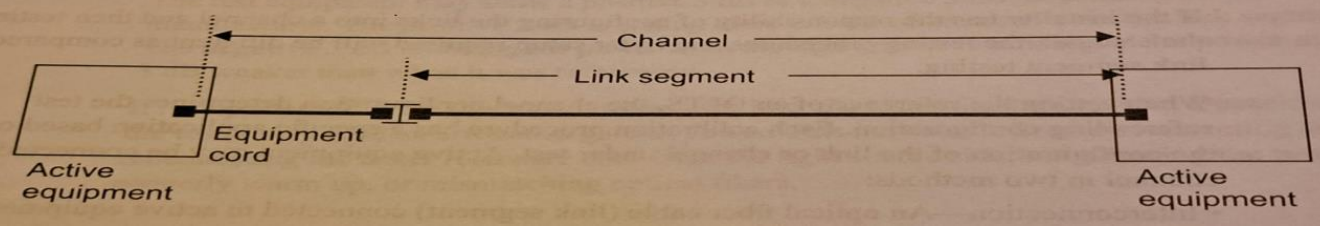


Figure 6.22
Dual interconnection

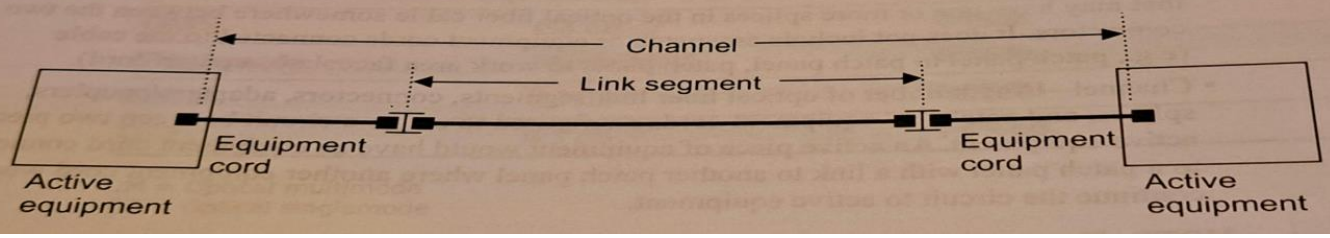
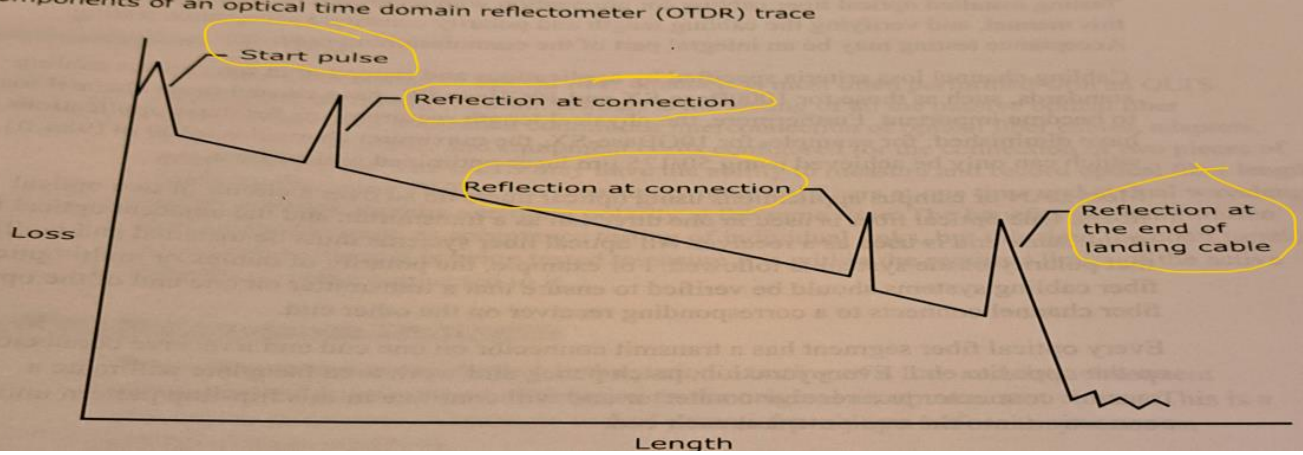
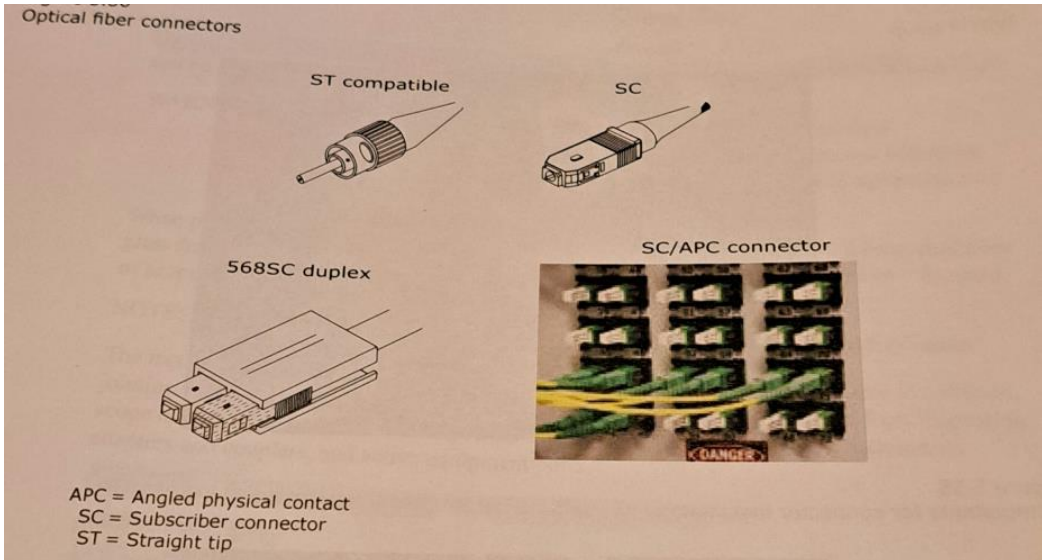
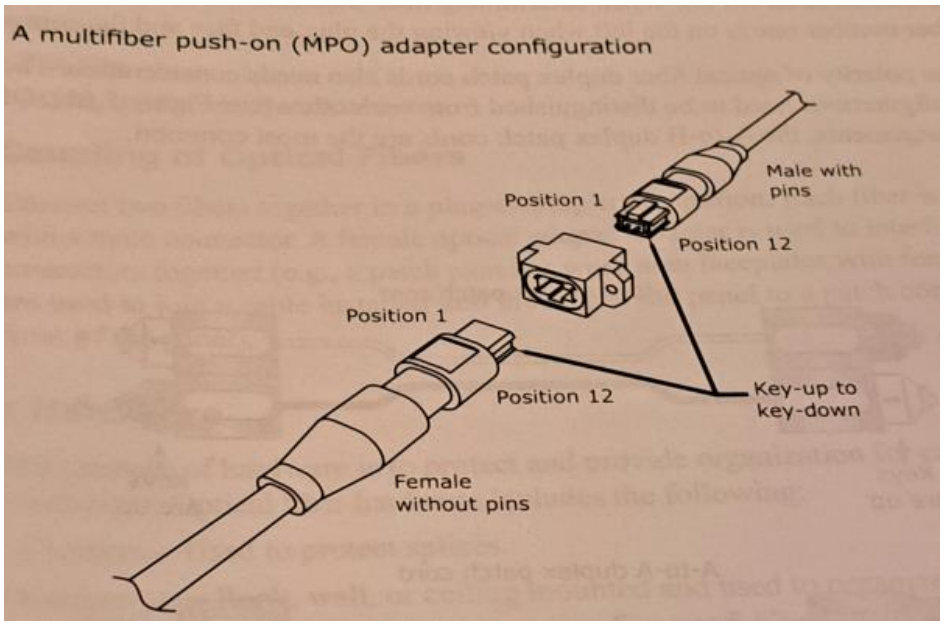


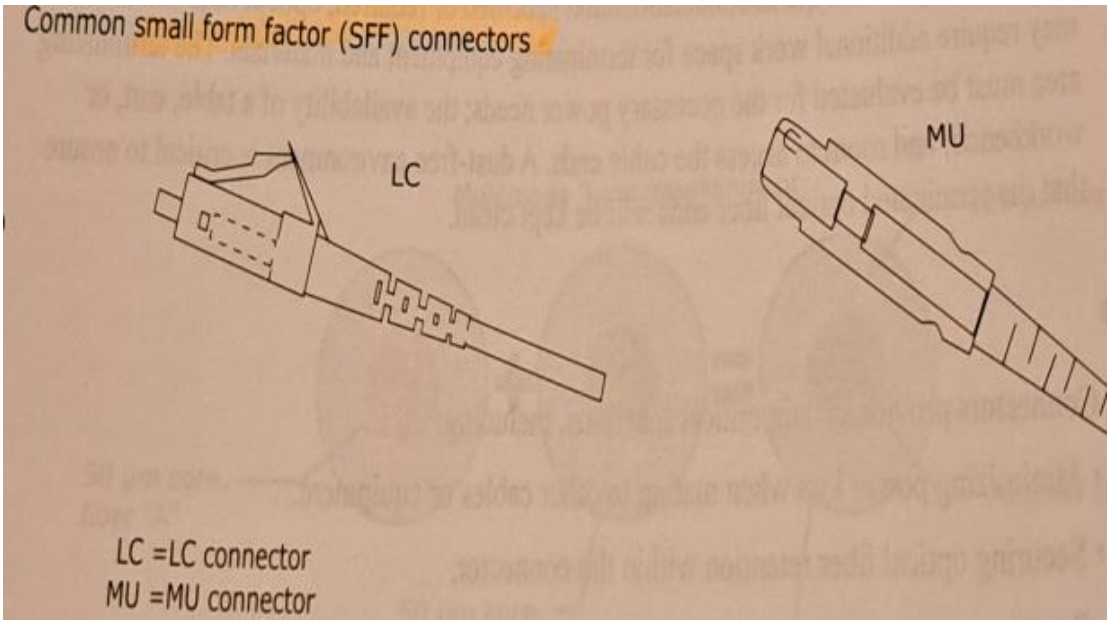
Figure 6.16
Components of an optical time domain reflectometer (OTDR) trace

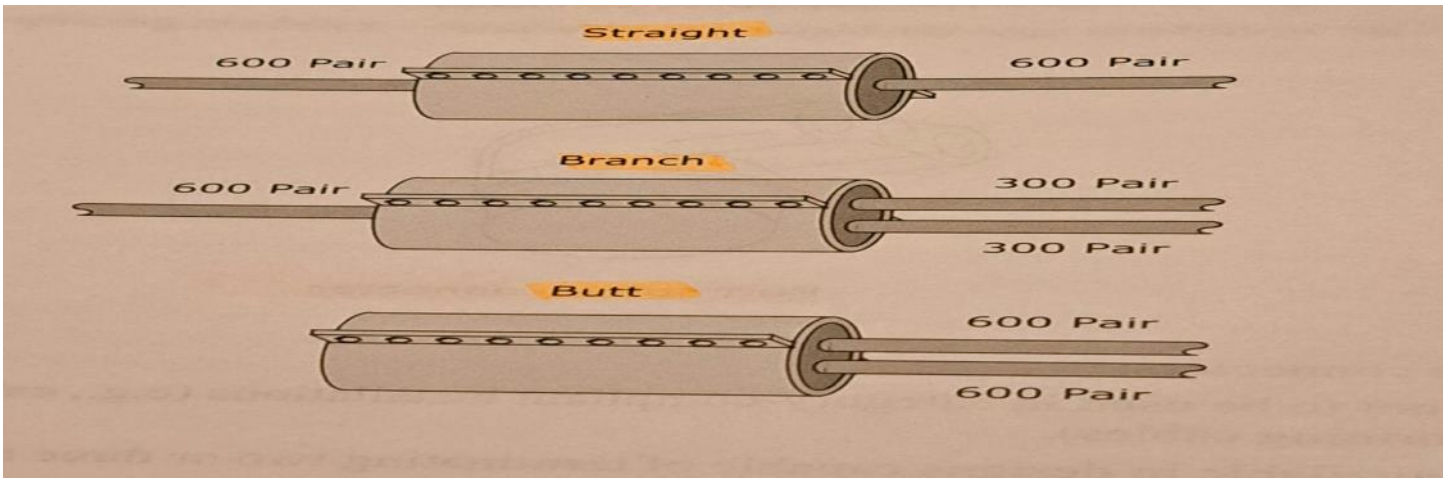


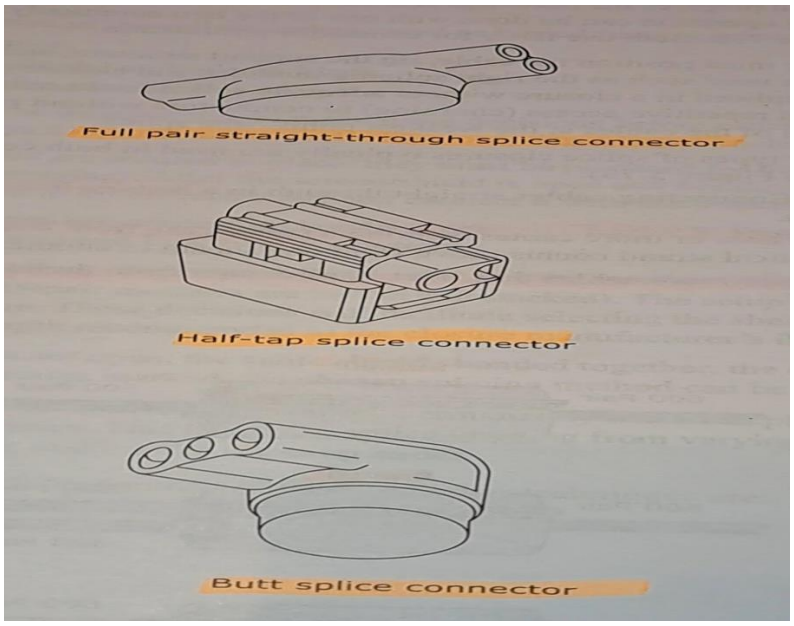


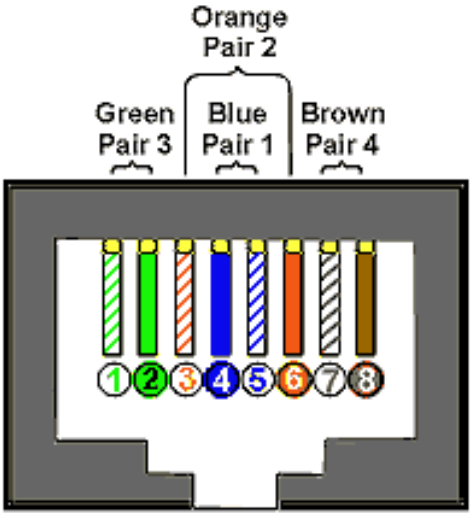
BICSI recommends that Table 5.5 be used for developing an administration solution.



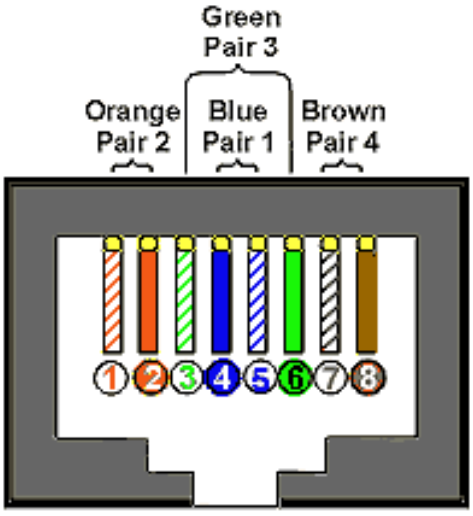






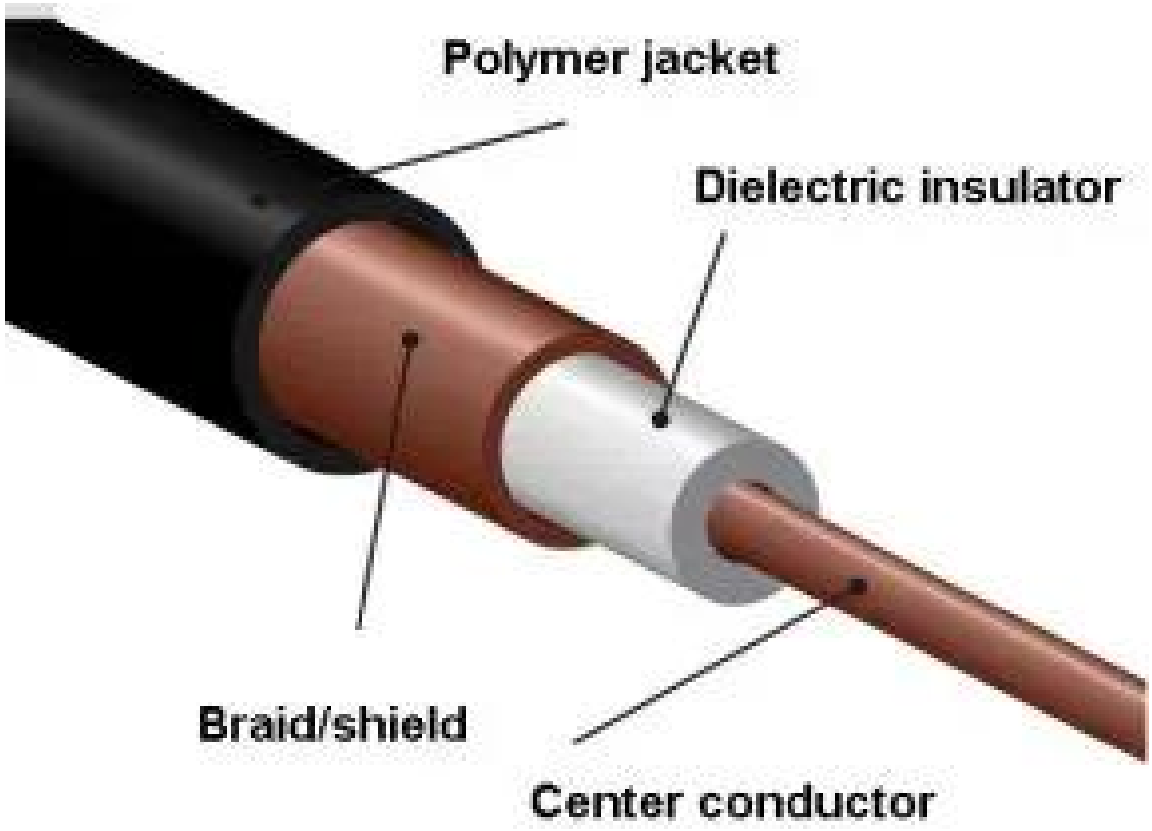


RJ-45 JACK
TIA/EIA 568A STANDARD



RJ-45 JACK
TIA/EIA 568B STANDARD

Table 5.5
Connector/adaptor color code

Fiber Type	Connector Color	Adapter Color	Cable Color
62.5/125 μm	Beige	Beige	Per manufacturer
50/125 μm	Black or beige	Black	Per manufacturer
50/125 μm laser enhanced	Aqua or beige	Aqua	Per manufacturer
Singlemode	Blue	Blue	Per manufacturer
Singlemode APC connector	Green	Green	Per manufacturer

APC = Angled physical contact